

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0704

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.
2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring

down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.
4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or

40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution

SOLUTIONS

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

1. Problem Analysis:

A 20-person startup requires a reliable, scalable, secure, and easy-to-manage computer network. The network must support approximately 20 employees along with printers, VoIP phones, laptops, and other network devices.

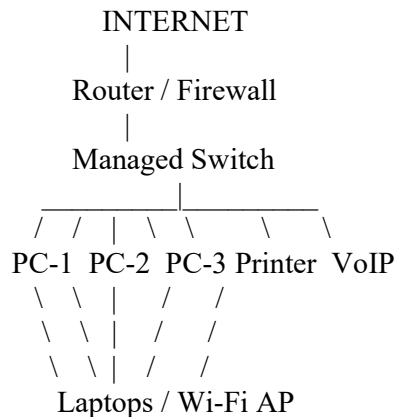
The main requirements are:

- Reliable connectivity for all employees.
- Easy troubleshooting and maintenance.
- High-speed communication between devices.
- Wireless connectivity for laptops and mobile devices.
- Internet access with basic security.
- Room for future expansion.
- Minimum impact when one device or cable fails.

Considering these requirements, a **Star Topology** is the most suitable network design because every device is connected to a central network switch.

2. Proposed Solution Design:

The proposed network uses a **24-port managed Layer 2/Layer 3 switch** as the central device. The basic architecture is:



All wired devices are connected individually to the central switch using **Cat6 UTP cables**. The switch is connected to a **router/firewall**, which provides internet connectivity, DHCP, NAT, and basic network security. A **wireless access point (AP)** is also connected to the switch to provide Wi-Fi connectivity.

A 24-port switch is selected instead of a 20-port switch because it provides additional ports for future expansion.

3. Technical Implementation:

Hardware Required

Device	Purpose
24-Port Managed Switch	Central connection point
Router/Firewall	Internet access, DHCP, NAT and security
Wi-Fi Access Point	Wireless connectivity
20 PCs/Laptops	Employee devices
Network Printer	Shared printing
VoIP Phones	Voice communication
Cat6 UTP Cables	Wired network connections

Network Configuration

Each workstation is connected directly to the central switch.

Example IP addressing scheme:

Network: 192.168.10.0/24

Gateway: 192.168.10.1

DHCP Range: 192.168.10.100 – 192.168.10.200

Subnet Mask: 255.255.255.0

The router/firewall acts as the default gateway and provides DHCP services. The switch manages communication between devices inside the local network.

For improved security, the network can be divided using **VLANs**, for example:

VLAN 10 – Employees

VLAN 20 – VoIP Phones

VLAN 30 – Guest Wi-Fi

VLAN 40 – Network Management

This improves security and reduces unnecessary network traffic.

4. Tools and Technologies:

The following tools and technologies can be used to implement and test the network:

- **Cisco Packet Tracer** – for designing and simulating the network.
- **Managed Layer 2/Layer 3 Switch** – for centralized connectivity.
- **Router/Firewall** – for internet access, DHCP, NAT and security.
- **Cat6 UTP** – for high-speed wired communication.
- **Wi-Fi Access Point** – for wireless connectivity.
- **Ping and IP configuration commands** – for connectivity testing.
- **VLAN technology** – for network segmentation and security.

Cisco Packet Tracer can first be used to simulate the complete network before physical implementation.

5. Problem-Solving Approach:

The solution follows a systematic approach:

Requirement Identification → Topology Selection → Device Selection → IP Addressing → Configuration → Testing → Troubleshooting → Future Expansion

Star topology was selected because it provides centralized management and excellent fault isolation.

For example, if the cable connecting PC-5 fails, only PC-5 loses connectivity. The remaining 19 employees continue to use the network normally.

This is better than a bus topology, where a major backbone failure could affect the entire network.

6. Innovation and Creativity

Although a basic star topology is suitable for the startup, additional features can make the solution more modern and scalable.

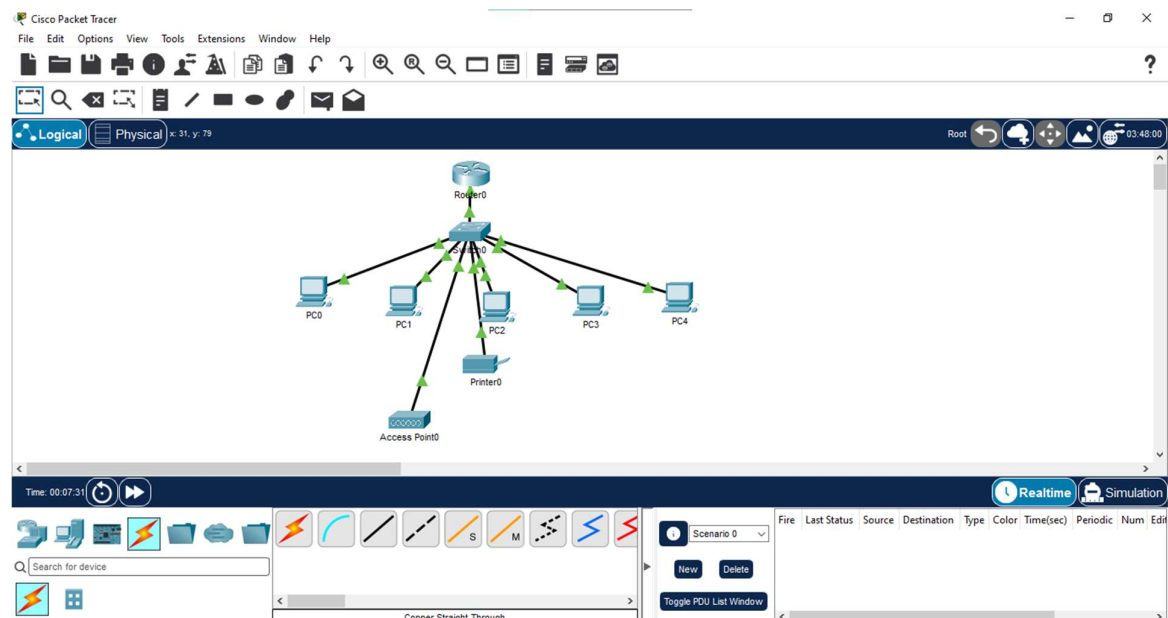
The proposed network can include:

- VLAN-based network segmentation.
- Separate guest Wi-Fi.
- QoS for VoIP traffic.
- Firewall-based access control.
- Centralized monitoring of network devices.
- Port security to prevent unauthorized devices.
- Future integration with cloud services.
- Backup internet connectivity for improved availability.

These improvements make the network suitable not only for the current 20 employees but also for future business growth.

7. Testing and Validation:

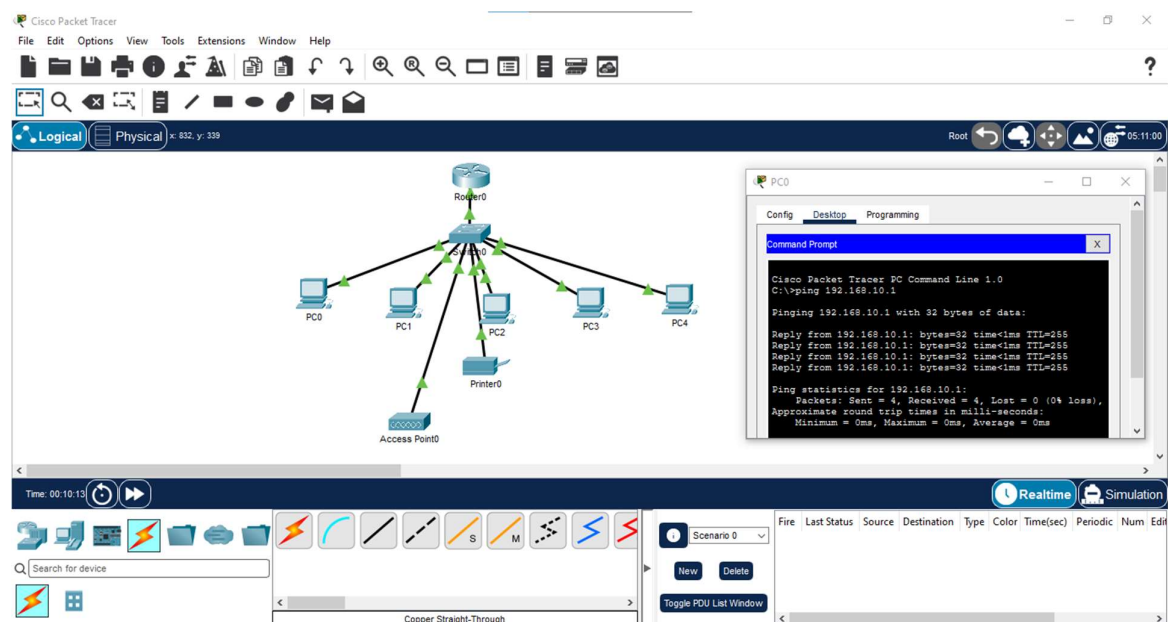
After implementation, the network should be thoroughly tested.



Test 1 – Connectivity Test

Use:

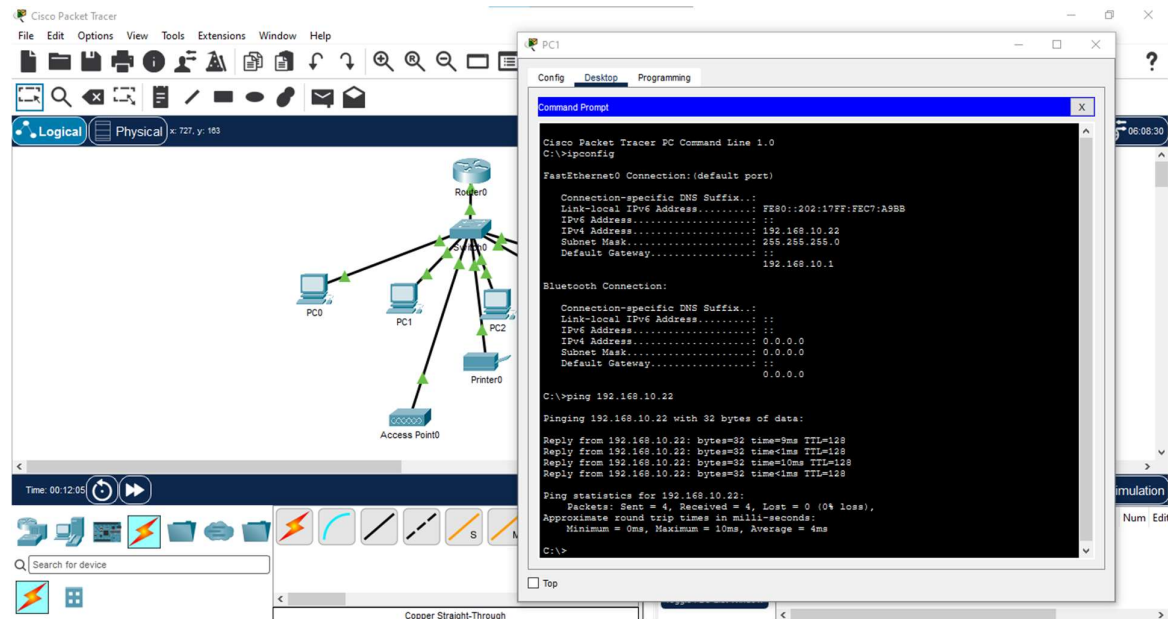
ping 192.168.10.1



This verifies connectivity between a workstation and the gateway.

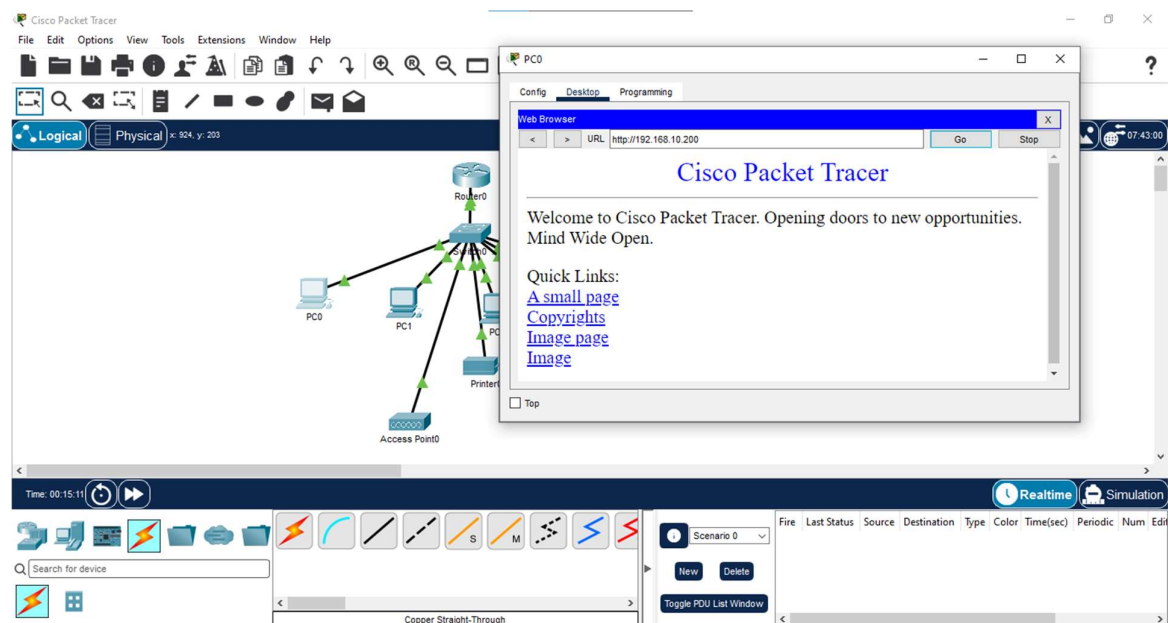
Test 2 – Device-to-Device Communication

ping 192.168.10.101



This checks communication between network devices.

Test 3 – Internet Connectivity



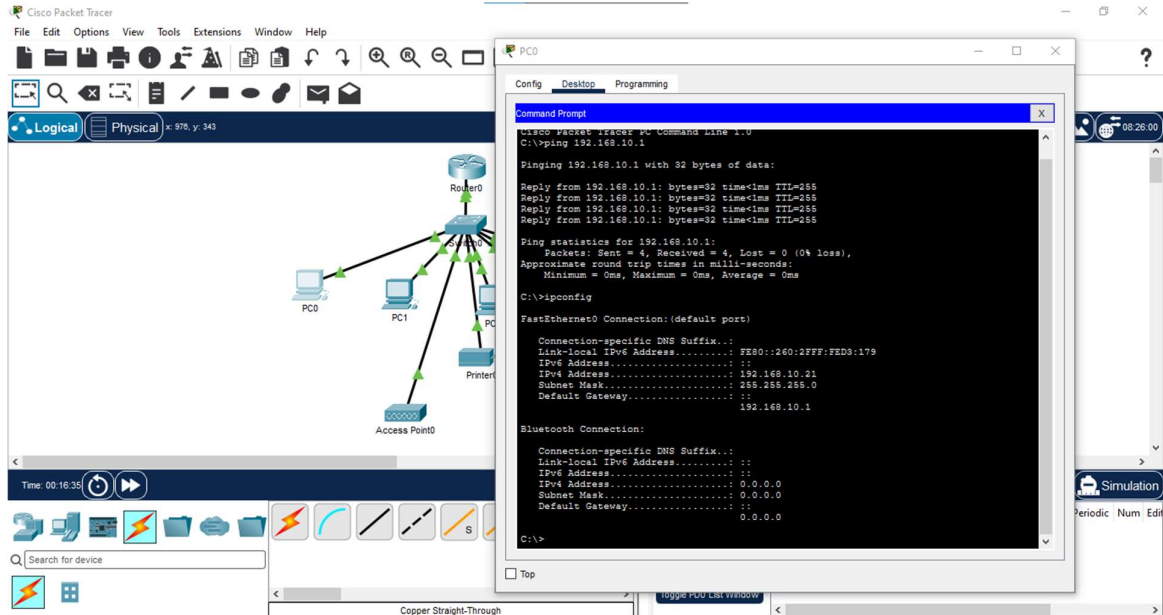
Open a website from a workstation and verify internet access.

Test 4 – DHCP Test

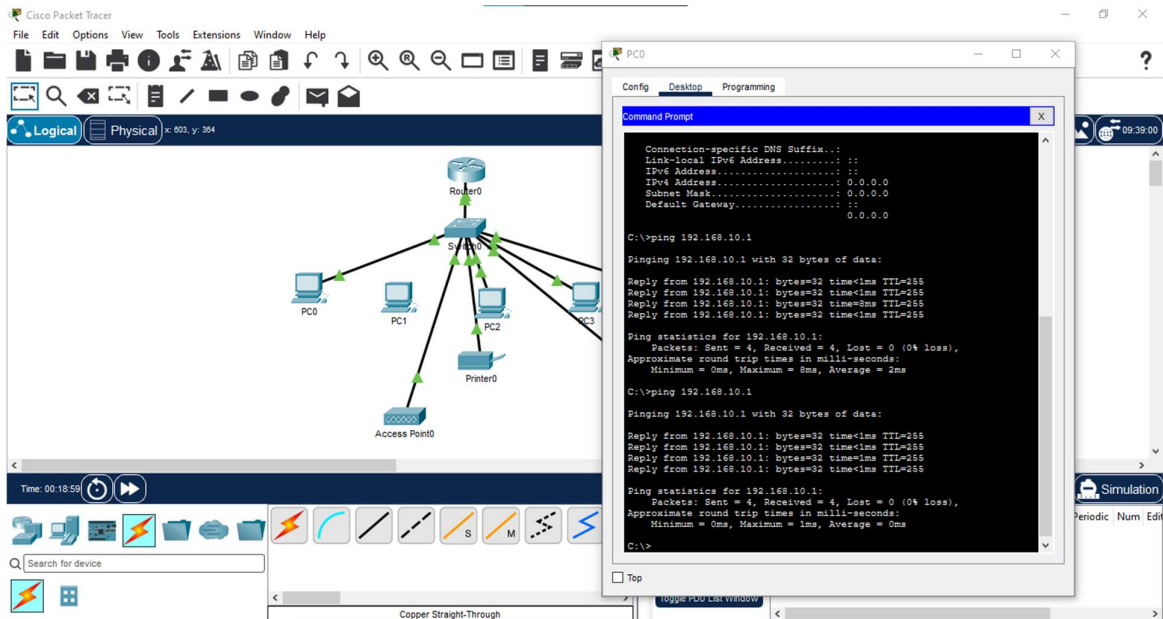
Check whether client devices automatically receive:

- IP address
- Subnet mask
- Default gateway

- DNS information



Test 5 – Fault Isolation



Disconnect one workstation's network cable and verify that only that workstation loses connectivity while the other devices remain operational.

Expected Result

The network should provide reliable communication, successful internet access, correct IP assignment, and effective fault isolation.

8. Advantages of the Proposed Design:

The star topology provides several important advantages:

1. **Easy management** – All connections are centralized through the switch.
2. **Easy troubleshooting** – Individual faulty cables or devices can be identified quickly.
3. **Fault isolation** – Failure of one workstation does not affect other users.

4. **Scalability** – Spare switch ports allow additional devices to be connected.
5. **High performance** – Dedicated connections provide good network performance.
6. **Security** – VLANs and switch security features can be implemented.
7. **Centralized control** – Network administrators can monitor and manage devices from the switch.

9. Conclusion:

For a 20-person startup, a **star topology with a 24-port managed switch, router/firewall, Wi-Fi access point, and Cat6 UTP cabling** provides a reliable, scalable, and manageable networking solution.

The design satisfies the company's current requirements while allowing future expansion. Centralized management makes troubleshooting easier, while individual device connections provide effective fault isolation. Additional technologies such as VLANs, QoS, firewall rules, and port security can further improve the network's performance and security. Therefore, the proposed star topology is an efficient and practical solution for the startup's networking requirements.